# Foundation Models for Image Segmentation

Traditionally, you had to train a custom computer vision model for every specific dataset. Foundation models eliminate this bottleneck. They apply general visual understanding to entirely new images. You can segment medical scans, satellite imagery, or everyday photos using a single model. You provide a prompt. The model instantly identifies and separates the objects.

| Approach | Core Concept | Your Data Requirements | Your Compute Cost | How You Apply It | Practical Example |
|---|---|---|---|---|---|
| **Training from Scratch** | You build and train a model entirely from zero. | You must collect and label massive datasets containing hundreds of thousands of images. | You will spend days or weeks running calculations on expensive GPU clusters. | You define the architecture and optimize every single network weight. | You train a custom neural network from zero to detect proprietary hardware defects in highly classified factory images. |
| **Transfer Learning** | You start with a model already trained on a general task. | You only need hundreds or thousands of labeled examples. | You finish training in minutes or hours using standard hardware. | You freeze the early network layers and retrain only the final output layers to recognize your specific classes. | You take a ResNet model trained on everyday objects and fine-tune it to classify specific types of skin lesions. |
| **Foundation Models** | You use a massive model pre-trained on vast | You need absolutely zero or very few labeled examples. | You pay zero training costs but incur higher inference compute | You guide the model instantly using direct text prompts, clicks, or | You use the Segment Anything Model (SAM) to instantly cut out |

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| | amounts of broad data. | | costs per request. | bounding boxes. | specific buildings in satellite imagery just by clicking on them. |

# Segment Anything Model (SAM) Working

Meta developed SAM as the first true foundation model for computer vision. SAM segments any object in any image. It requires zero custom training for new tasks.

You interact with SAM using direct prompts. You click on a target, draw a bounding box around it, or write a text description. SAM instantly generates a precise cutout mask. SAM handles ambiguity effortlessly. If you click a person's shirt, the model might interpret your intent as just the shirt or the entire person. To resolve this, SAM simultaneously outputs multiple valid masks. You simply select the exact boundary you need.

# SAM Architecture

SAM achieves its remarkable speed and flexibility through a strictly decoupled three-part architecture.

- **Image Encoder:** You feed your image into this massive component once. It uses a Vision Transformer (ViT) to extract a dense mathematical representation of the image. This step is computationally expensive but only happens a single time per image.
- **Prompt Encoder:** You provide your interactive inputs here. It translates your clicks, boxes, or text into lightweight embeddings. This component runs instantly.
- **Mask Decoder:** This component combines the heavy image features and your lightweight prompt embeddings. It maps this combined data to final segmentation masks. It runs in milliseconds. You get real-time, interactive segmentation directly in your web browser.

# Segment Anything Model (SAM) Architecture

Hover or tap components to inspect input flow between heavy one-time processing and real-time prompting.

